

MA dot dash

- Scope
 - In Scope
 - See below
 - Out of Scope
 - Optimization of process
 - Experiments
 - Upscaling
- Reference case introduction (e.g., incumbent BASF process (for detailed comparison in module 3) or S&P PEP sheets, Process Economics Program (for high level economic evaluation in module 2, possible also for all endothermic high temperature processes that could potentially be replaced))
 - Potential processes to be replaced
 - A
 - B
 - C
- Potential of new technology (what are thermodynamical/physical limitations of the process? Basic feasibility)
 - Optimal reaction conditions
 - All reaction pathways (full reaction set)
 - Constraints (Development feasibility)
 - Competitive landscape (e.g., does the idea exist already?)
 - Technology readiness for the different steps of the process (i.e., is there any part of the process for which technology does not exist?)
 - Materials of construction constraints
 - Fabrication or construction constraints
 - The development and time-to-market timeframe requirement
 - Appendix A Beretta
 - Task 1 - Development of a Block Flow Diagram (BFD) for the process idea
 - CO₂ input at cost of negative carbon price, purification unit CO₂, novel reactor, product preparation
 - No purification needed, incumbent reactor, synthesis gas ratio manipulation, product preparation
 - Task 2 - Evaluation of reference cases (i.e., competitive industrial processes)
 - Task 3 - High level economic evaluation for the developed BFD
 - Task 4 - Review of industrial process product specifications
 - Task 5 - High level Safety Health and Environmental (SHE) review of the process
 - Task 6 - Evaluation of the intellectual property (IP) position and landscape
- Techno-Economic Assessment
 - Developing an expanded flowsheet for the process technology
 - requires significant judgement which unit operations to choose for peripheral process steps around tovel technology
 - BASF flowsheets for reference case are needed
 - Detailed mass and energy balance for the developed flowsheet
 - Sizing of the process equipment involved in the process

- Estimating the operating expenses (OPEX) and capital expenses (CAPEX)
 - CAPEX (Appendix D Beretta for more details, or Seider (4) section 16)
 - Fixed capital investment
 - Purchased equipment cost
 - Equipment list from flowsheet
 - Sizing (potentially with Aspen Plus or HYSYS)
 - Function of purchased equipment cost estimate
 - Various direct costs
 - Various indirect costs
 - Working capital investment
 - OPEX (A more detailed approach with a more comprehensive breakdown is provided by Seider (4))
 - Variable cost
 - From Flowsheet and costs
 - Raw materials
 - Utilities
 - first-pass electrical power consumption based on preliminary sizing of the equipment
 - Effluent treatment
 - Function of cost of CAPEX estimate
 - Operating supplies
 - Fixed costs
 - Cost of labor estimate
 - Function of cost of labor estimate
 - Supervision
 - Plant overheads
 - Function of cost of CAPEX estimate
 - Maintenance
 - Taxes and insurance
 - Financing costs
 - General expenses
 - Function of variable and fixed costs
 - Completing an economic analysis of the process
 - Case 1: Construction of complete new process plants
 - Case 2: Retrofitting existing plants
 - Identifying and summarizing process risks that must be addressed
 - Sensitivity analysis for assumptions
- Optional: LCA (a lot of data already at hand)